

INTERNAL REFERENCE

TradeFlow

Who does what: a role-by-role guide to the Platform Owner, the Admin, and the User, for the shipment, client, and duty management system built for Ghanaian traders and freight forwarders.

Covers the application as of Phase 5 (hardening & compliance) plus team invites (multi-user organizations).
Three roles, one platform: Platform Owner (runs the deployment) · Admin (maintains the shared duty schedule) · User (day-to-day trader / forwarder staff).

1. Platform Owner

RUNS THE WHOLE DEPLOYMENT

The Platform Owner is whoever controls the infrastructure TradeFlow runs on: the Supabase project (database, auth, file storage) and the Vercel deployment. This is an operational and legal role, not something you sign up for inside the app — it's whoever holds the hosting accounts and environment variables. Today that's a single person; at scale it might be a small ops team.

Initial setup

1. Create a Supabase project and copy its Project URL and anon public key.
2. Deploy the codebase to Vercel (recommended host — see the note at the end of this section) and set four environment variables: `NEXT_PUBLIC_SUPABASE_URL` , `NEXT_PUBLIC_SUPABASE_ANON_KEY` , `DATABASE_URL` , and `ADMIN_EMAILS` .
3. Apply seven database migrations, in order. Four of them **only run through the Supabase SQL Editor** — they are not part of the automated migration command:

File	What it does	How it's applied
<code>0000_lonely_mauler.sql</code>	Tables, enums, foreign keys	<code>npm run db:migrate</code>
<code>0001_rls_and_triggers.sql</code>	Row-level security policies; auto-creates an organization + profile on signup	Manual (SQL Editor)
<code>0002_notifications.sql</code>	The notifications table	<code>npm run db:migrate</code>
<code>0003_documents_storage.sql</code>	Private document storage bucket + its access policies	Manual (SQL Editor)
<code>0004_profile_column_grants.sql</code>	Closes a privilege-escalation gap in Supabase's default API grants	Manual (SQL Editor) — not optional
<code>0005_invites_table.sql</code>	The invites table	<code>npm run db:migrate</code>
<code>0006_invites_rls_and_join_trigger.sql</code>	Invites RLS; updates the sign-up trigger to join an inviting organization when a match exists	Manual (SQL Editor)

4. In the Supabase dashboard, keep **Auth** → **Confirm email** switched on. This is the actual security boundary behind the Admin allowlist below — see the warning under Admin.

Ongoing responsibilities

- **Manage the `ADMIN_EMAILS` list** — a comma-separated list of email addresses granted the platform-wide Admin capability described in section 2. This is the only access-control decision the Platform Owner makes directly.
- **Review the audit log** — every client, shipment, document, and tariff change is recorded (who, what, when). Useful for security review and for demonstrating accountability under data protection law.
- **Data protection compliance** — Ghana's Data Protection Act (Act 843) requires registering as a data controller with the Data Protection Commission, an actual privacy policy (the app does not have one published yet — this needs real legal content, not placeholder text), and a decision on how long shipment/customs records are retained versus a customer's right to request erasure. These are organizational decisions, not settings inside the app.
- **Backups and monitoring** — Supabase manages database backups on paid tiers; confirm the plan matches your recovery needs.

Multi-user organizations: a sign-up only creates a brand-new organization when there's no pending invite waiting for that email (see *Team* in section 3). The Platform Owner doesn't need to do anything to enable this — any org member can invite a teammate themselves.

Recommended host: Vercel. The codebase uses very recent Next.js App Router conventions (server actions, edge middleware, a custom security-headers config) that Vercel — built by the Next.js team — supports with zero configuration. A generic Node host works too but loses that day-one alignment.

2. Admin (Tariff Administrator)

PLATFORM-WIDE, NOT TIED TO ANY ONE ORGANIZATION

An Admin is simply a signed-up User whose login email appears in the Platform Owner's `ADMIN_EMAILS` list. It is a narrow, single-purpose role: maintaining Ghana's official customs duty rate schedule, which is *shared reference data* used by every organization's duty calculator — not a per-organization owner or manager role.

What an Admin can do

At `/dashboard/admin/tariffs`, visible only to allowlisted emails:

- View every HS code / duty rate entry currently published
- Add a new entry: HS code, description, rate percentage, effective date
- Edit an existing entry
- Delete an entry — blocked automatically if that rate has already been used in a real duty calculation, to keep historical shipment records accurate

What an Admin cannot do

Nothing beyond the ordinary User capabilities described in section 3. Being on the Admin allowlist does not grant visibility into any organization's clients or shipments, and does not make someone an owner or manager within any organization — those are two completely separate concerns in this system.

Security note for the Platform Owner: the Admin allowlist checks a user's *email* against `ADMIN_EMAILS`. That check only holds if Supabase's "Confirm email" setting stays enabled — otherwise someone could sign up claiming an allowlisted email address without proving they own it. Never disable email confirmation on this project.

Demo data: running `npm run db:seed` inserts a handful of illustrative rates, each description prefixed `[EXAMPLE]`. These are not real Ghana Revenue Authority rates. An Admin should replace them with the actual published schedule before anyone quotes a real client.

3. User (Trader / Freight Forwarder Staff)

EVERYDAY USE, SCOPED TO YOUR OWN ORGANIZATION

Anyone can sign up. Unless a teammate already invited your email (see *Team* below), signing up automatically creates a new organization (named after you) and a profile inside it — there's nothing to configure before you start working.

Getting started

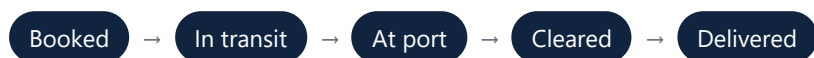
1. Go to **/signup**: enter your full name, email, and a password (minimum 8 characters — use the eye icon to check what you typed before submitting).
2. Confirm your email from the message Supabase sends you — you cannot log in until this step is done.
3. Log in at **/login**. You'll land on the Overview dashboard: active shipment count, client count, and duty calculated this month.
4. Optional: install TradeFlow to your phone or desktop home screen. A prompt appears automatically in supported browsers; on iPhone/iPad, use Share → Add to Home Screen.

Clients

/dashboard/clients — add, edit, and delete the businesses you ship for (name, email, phone). A client can't be deleted while it still has shipment history attached; remove or reassign those shipments first.

Shipments

/dashboard/shipments — the core of the app. Creating a shipment asks for a client, HS code, description, customs value, and quantity, and shows a live estimated duty as you type, drawn from the tariff schedule an Admin maintains. Each shipment moves through a fixed status pipeline:



Editing a shipment lets you change any field, including status, and attach supporting documents.

Duty calculator

/dashboard/calculator — a standalone version of the same estimator, for a quick one-off check without creating a shipment record. If an HS code has no published rate yet, it says so plainly rather than guessing.

Documents

Attach files to any shipment from its edit panel: invoices, packing lists, certificates of origin, or other supporting paperwork. Accepts PDF, PNG, JPEG, or WebP, up to 10 MB each, stored privately — only members of your own organization can open them.

Notifications

The bell icon shows in-app alerts whenever a shipment's status changes, sent to every other member of your organization (not the person who made the change). Email notifications are not wired up yet — alerts are in-app only for now.

Analytics & export

/dashboard/analytics — all-time totals, six-month shipment-volume and duty-calculated trend charts, your top HS codes and top clients by activity, and a one-click CSV export of every shipment for use in Excel or Sheets.

Document type	Label shown in the app
invoice	Invoice
packing_list	Packing list
certificate_of_origin	Certificate of origin
other	Other

Team

/dashboard/team — invite a colleague to join your organization by email, from a button any member can use. There's no separate acceptance step or invite link to click: the first time they sign up with that exact email, they land in your organization automatically instead of getting a fresh one of their own. From that point on they see the same clients, shipments, documents, and notifications you do.

- Pending invites (not yet accepted) can be revoked from the same page.
- Inviting an email that already has a TradeFlow account is rejected — there's no way yet for an existing user to switch organizations.
- Everyone in an organization currently has equal, full access. The role shown next to each member's name (Owner for whoever signed up first, Agent for everyone invited in) is informational only for now — it doesn't restrict what anyone can do yet.

Roles at a glance

Role	Scope	How you get it	Key capability
Platform Owner	The whole deployment	Holds the Supabase/Vercel accounts	Infrastructure, environment config, the Admin allowlist, compliance decisions
Admin	Platform-wide (all organizations)	Email listed in <code>ADMIN_EMAILS</code>	Maintain the shared customs duty rate schedule
User	Own organization only	Sign up (or accept an invite by signing up with the invited email)	Clients, shipments, duty estimates, documents, notifications, analytics, inviting teammates

Known limitations, honestly stated

- Roles (Owner/Agent/Viewer) are stored but not enforced — every organization member has equal full access. Real per-role restrictions are a separate, future change.
- No way for an existing TradeFlow user to switch or join a second organization — invites only work for brand-new sign-ups.
- Notifications are in-app only; no email delivery yet.
- No self-service account deletion, and no published privacy policy or terms of service.
- Formal Data Protection Commission registration is an organizational step for the Platform Owner, not something the app does automatically.

TradeFlow — internal usage reference. Generated to accompany the Phase 5 hardening pass and the team-invites feature; reflects the application's actual behavior as implemented, not aspirational features.